MILITARY SPECIFICATION

DARKROOM, PHOTOGRAPHIC, PORTABLE ES-38( )

This specification is approved for use by Electronics Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the installation of components in Shelter, Electrical Equipment S-280( )/G, to form one type of truck transportable photographic darkroom designated as Darkroom Photographic, Portable ES-38( ) (see 6.4).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-F-475 - Floor Covering Vinyl, Surface (Title And Roll), With Backing.
L-P-375 - Plastic Film, Flexible, Vinyl Chloride.
V-T-285 - Thread, Polyester.
QQ-S-766 - Steel Plates, Sheets, And Strip-Corrosion Resisting.

MILITARY

MIL-P-116 - Preservation, Methods Of.
MIL-M-13231 - Marking Of Electronic Items.
MIL-F-14072 - Finish For Ground Electronic Equipment.
MIL-A-14372 - Air Conditioner, Base-Mounted, Self-Contained Type, Air-Cooled, Nominal, 9,000 Btu/hr.
MIL-P-26767 - Printer, Contact, Photographic, Continuous EN-6B.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Research and Development Command, ATTN: DELCS-PE, Fort Monmouth, NJ 07703-5304, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) Form1426) appearing at the end of this document or by letter.

FSC 6780

Source: https://assist.dla.mil -- Downloaded: 20260-26T04:14Z
Check the source to verify that this is the current version before use.

MIL-P-55157	- Printer, Projection, Photographic EN-36( ).
MIL-P-55159	- Processing Machine, Photographic Film And Paper EH-29( ).
MIL-S-55286	- Shelter, Electrical Equipment S-280( )/G.
MIL-S-55507	- Shelter, Electrical Equipment (With Or Without Equipment), Packaging Of.
MIL-P-55594	- Printer, Contact, Photographic EN-100( ).

STANDARDS

FEDERAL

FED-STD-751	- Stitches, Seams, And Stitchings.

MILITARY

MIL-STD-105	- Sampling Procedures And Tables For Inspection By Attributes.
MIL-STD-252	- Classification of Visual And Mechanical Defects For Equipment, Electronic, Wired, And Other Devices.
MIL-STD-454	- Standard General Requirements For Electronic Equipment.
MIL-STD-461	- Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement Of.
MIL-STD-810	- Environmental Test Methods.
MS27130	- Nut, Plain, Blind Rivet, Flat and Countersunk Head, Open End.

2.1.2 Other Government documents. The following Government document form a part of the specification to the extent specified herein.

DRAWINGS

ELECTRONICS COMMAND

SC-D-82768	- Aircraft Loading Data Plates For Trailer, Semi-Trailer, Shelter Truck And Equipment.

US AIR FORCE

49D9763	- Bracket Assembly-Universal Film Spool.
49D18214	- Table Assembly-Film Plotting, 9x18 in. Negative, Type A-7.

(Copies of specifications, standards, and drawings required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specfication to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70-78 - National Electrical Code - 1978.

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210).

(Industry assocation specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The darkroom is a complete self-contained mobile photographic laboratory installed in an S-280( )/G electrical equipment shelter. The capabilities of the darkroom include continuous automatic processing for 70mm, 5 in and 9.5 in roll film; 9.375 in roll paper; automatic step and repeat and continuous contact printing; and automatic projection printing.

3.1.1 Major components. The major components of the E-38( ) photographic darkroom include the following:

Shelter, electrical equipment S-280( )/G (see 3.5).
Heaters and air conditioner (see 3.8).
Processing machine, photographic film and paper EH-29( ) (see 3.9.1).
Printer, projection, photographic EN-36( ) (see 3.9.2).
Printer, contact, photographic EN-100( ) (see 3.9.3).
Printer, contact, photographic, continuous EN-6( ) (see 3.9.4).
Plotting table, photographic film AR-92( ) (see 3.9.5).

3.2 First article. The quantity of samples specified in the contract shall be subjected to first article inspection (see 4.3 and 6.2b).

3.3 Standard sample. A standard sample of the darkroom will be available for inspection by prospective bidders and when specified (see 6.2c), will be lent to the contractor. Unless otherwise specified (see 6.2d), physical construction of the darkroom shall conform to the standard sample. (NOTE: In case of conflict between specified performance characteristics for the equipment and the performance of the model, the specified performance characteristics govern.)

3.4 Material. The material shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the darkroom to meet the requirements of this specification. Acceptance or approval of any constituent material shall not be contrued as guaranty of the acceptance of the finished product.

3.4.1 Recycled material. It is encouraged that recycled material be used, when practical, provided it meets the requirements of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

3.4.2 Tropicalization.

3.4.2.1 Treatment of materials. Fungus-inert materials shall be used where practicable. These materials shall be in accordance with requirement 4 of MIL-STD-454. When materials are required to be treated, they shall be cleaned in accordance with requirement 9 of MIL-STD-454 and then shall be treated in accordance with 3.4.2.2 through 3.4.2.5 herein.

3.4.2.2 Treating compounds. Treating compounds containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating compound is compatible with the material or surface to be treated. Selection of the treating compound shall be such that there shall be no increase in flammability of the treated material.

3.4.2.3 Toxicity. Treating compounds shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for purpose intended.

3.4.2.4 Flexibility. Treatment shall not affect the flexibility of the treated material to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.4.2.5 Statement of treatment. The statement of treatment shall indicate whether or not nutrient materials will be used. If nutrient materials will be used, the statement shall also describe in detail the materials to be treated and the treating compounds and processes that the contractor proposed to use. (See 6.2e).

3.5 Shelter. Shelter, electrical equipment S-280( )/G conforming to MIL-S-55286, shall be used to house the equipment required for the darkroom.

3.5.1 Film port. A watertight and light-tight film port shall be installed in the shelter door above the fording line in accordance with the standard sample. Provisions shall be provided to prevent the film from dropping onto the floor. The film port shall be capable of accomodating magazine, film, cassette type LA-135( ), and LA-414( ).

3.5.2 Jack pads. Jack pads shall be provided to permit leveling of the shelter. The pads shall be installed in accordance with the standard sample.

3.5.3 Interior painting. The shelter interior, four walls and ceiling, shall be painted with a chemical and moisture resisting enamel paint consisting of 4 parts of Sherwin-Williams white, type B69W12 to 1 part of Sherwin-Williams green, type B69G19.

3.5.4 Shelter openings. Insofar as possible, all shelter openings shall be located above the "fording line".

3.5.5 Floor covering. After the walls have been painted as specified in 3.5.3 above, the floor plus the lowest 6 inches (152mm) of each wall shall be covered with vinyl sheet conforming to L-F-475, type II, grade A. After installation of equipment, all mounting points shall be sealed with a chemical resistant sealant.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

3.6 Plumbing system. A plumbing system shall be installed in the shelter in accordance with the standard sample.

3.6.1 Piping Pressure. All piping joints and connections shall be capable of withstanding a pressure of 90 lbs/in^2 (psi) (620kPa) without leakage. Stainless steel pipe shall be used in all hot water lines.

3.6.2 Sink. A 12 by 10 inch (305 by 254mm) sink shall be provided. The sink shall be stainless steel conforming to QQ-S-766, class 316. The sink shall have a water basin type funnel drain at the bottom center and shall have rolled top edges.

3.6.3 Drainage. With the shelter in the level position and the discharge end of a 50 foot (15m), 0.750 inch (19mm) inside diameter (ID) hose no more than 2 feet (0.6m) lower than the floor of the shelter, drainage by gravity of all water, chemical solutions, etc., shall be completed in not more than 15 minutes.

3.6.4 Pump. The pump shall be a series BF, 0.5 HP, 120 volt water pump, as supplied by Gould Pumps, Inc. Seneca Falls, New York, or equal, and shall be capable of maintaining a total constant discharge of 3 gallons (11 liters) per minute from the hot and cold water faucets while pumping from a source located 20 feet (6m) below and 100 feet (30m) from the pump through a 0.750 inch (19mm) ID heavy duty flexible hose.

3.6.5 Hot water heater. The hot water heater shalll be a gasoline burning 60,000 BTU/hr (17.50kw) heater, Hunter Model UH-58-SC-1AW, 115 volt, 60 Hertz (Hz), single phase, or equal, and shall be installed in accordance with the standard sample.

3.6.6 Pressure regulator. An adjustable pressure regulator shall be provided and shall be capable of reducing commercial water pressure to 40 psi (276kPa).

3.6.7 Hot water thermostat. An adjustable thermostat shall be provided for controlling the temperature of the hot water within $\pm$2°F ($\pm$1.1°C) over a range of 95°F (35°C) to 180°F (82°C).

3.6.8 Liquid level switch. An automatic liquid level switch shall be provided and shall be capable of shutting off the hot water heater in case the water level drops below the top of the combustion tube.

3.6.9 Pressure relief valve. A pressure relief valve shall be provided on the hot water tank and shall be capable of releasing the pressure between 75 psi (517kPa) and 85 psi (586kPa).

3.7 Electrical system. An electrical system shall be installed in the shelter in accordance with the standard sample and shall enable the equipment to operate from a 120/208 volt $\pm$10 percent, 60 Hz $\pm$5 percent, 3 phase, 4 wire power source. The power consumption shall not exceed 15 kilowatts.

3.7.1 Color coding. All electrical power cables shall be color coded in accordance with National Electrical Code NFPA 70-78, Article 210-5.

3.7.2 Power cable connector. The power cable connector fitting shall be equipped with a Pyle National ZZM-WL-2016-2250 waterproof seal, or equal, and potted with potting compound General Electric Type RTV-11, or equal. After installation, the connector shall have a 10 megohm resistance when tested as specified in 4.6.2.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

3.7.3 Alarm. A buzzer circuit shall be provided such that the buzzer will sound when the external button is depressed regardless of what mode of operation the darkroom is operating.

3.7.4 Breakdown voltage, insulation. There shall be no flashover or breakdown when the equipment is tested as specified in 4.6.2.1. All adjacent metal parts not in circuit under test shall be considered as ground.

3.8 Heating and air conditioning.

3.8.1 Space heater. A hot air space heater, Hunter Model UH-48, or equal, shall be installed in the shelter as shown in the standard sample.

3.8.2 Air conditioners. Two air conditioners, conforming to MIL-A-14372, shall be installed in the shelter as shown in the standard sample. Canvas covers shall be provided to enclose the air conditioners when not in use.

3.9 Film and paper processing equipment.

3.9.1 Processing machine, photographic film and paper EH-29( ). A photographic film and paper processing machine conforming to MIL-P-55159 shall be installed in the shelter in accordance with the standard sample.

3.9.1.1 Mounting. Processing machine, photographic film and paper EH-29( ) shall be so mounted to permit movement of the processor for servicing and cleaning beneath the equipment. Materials utilized shall be corrosion resistant to the chemicals used in processing.

3.9.2 Printer, projection, photographic EN-36( ). Printer, projection, photographic EN-36( ), conforming to MIL-P-55157 shall be installed in the shelter in accordance with the standard sample.

3.9.3 Printer, contact, photographic EN-100( ). Printer, contact, photographic EN-100( ), conforming to MIL-P-55594, shall be installed in the shelter in accordance with the standard sample.

3.9.4 Printer, contact, photographic, continuous EN-6( ). Printer, contact, photographic, continuous EN-6( ), conforming to MIL-P-26767, shall be installed as shown in the standard sample.

3.9.5 Plotting table, photographic film AR-92( ). Plotting table, photographic film AR-92( ), consisting of Table, plotting film type A-7 in accordance with Air Force Drawing 49D18214 and modified in accordance with the standard sample, shall be installed in the shelter as shown in the standard sample.

3.9.6 Mixer-distributor, photographic chemical solution. Mixer-distributor, photographic chemical solution, consisting of Jet Mixer portable, Oscar Fisher Model JM-10, or equal, modified to pump both to and from the processing tanks, shall be installed as shown in the standard sample. The undercarriage of the mixer shall be provided with 4 casters and the handle shall be located at a position opposite to the pump to improve stability.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

3.10 Film and paper rewinder. A hand operated film and paper rewinder assembly consisting of a bracket assembly, in accordance with Air Force Drawing 49D9763, mounted on a board 10 by 15 by 0.75 inches (254 by 381 by 19mm) shall be provided. The rewinder assembly shall be capable of handling 100 foot (30m) lengths of 70mm, 5 inch and 9.50 inch film and 200 foot (61m) lengths of 9.375 inch paper. The assembly shall be stored on the front shelter wall.

3.11 Dust cover. Dust covers shall be provided to cover the following equipment installed in the shelter and shall be easily removed.

Processing machine, photographic film and paper EH-29( ).
Printer, projection, photographic EN-36( ).
Printer, contact, photogrphic EN-100( ).
Printer, contact, photographic, continuous EN-6( ).

3.11.1 Cover material. The cover shall be vinyl plastic conforming to L-P-375, type II, class 1, 0.008 inch (0.20mm) thick. Seams shall be in accordance with FED-STD-751, type 301, class SSc. Thread shall conform to V-T-285.

3.12 Shelves, cabinets, brackets, and supports. Shelves, cabinets, brackets, and supports shall be fabricated and installed in accordance with the standard sample. The brackets and support shall not be attached to the skin of the shelter only but shall be firmly attached to the mounting members.

3.13 Light-tight container. A light-tight container shall be provided and installed in the shelter in accordance with the standard sample to prevent photograph film from being light struck.

3.14 Securing of parts. Brackets, lugs, flanges, inserts, bolts, and other mounting arrangements shall be such as to retain items securely when subjected to specified service conditions. Parts shall be so mounted that loosening, wear of mounting arrangements, or permanent separation of parts will not occur when the items in which they are used are subjected to specified bounce test. Any permanent deformation resulting from the test shall not cause degradation of performance below full specification requirements. When practicable, friction between mating surfaces shall not be employed as the sole means of preventing fixed parts from rotating. Parts shall be secured in such a manner that failure of a single rivet or screw will not free the part completely.

3.14.1 Riveting. Riveting shall not be used for mounting parts such as brackets, wire way, etc., to only the skin of the shelter. Thickness of countersunk metal which accepts the heads of flush rivets shall be not less than the height of the rivet heads. The distance from the centers of rivet holes to the edges of the material in which the holes are formed shall be at least 1.5 times the rivet diameter. After riveting, the joints shall be tight, the jointed parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces.

3.14.2 Threaded inserts. No sheet metal or self-threading screws shall be used on either the inside or outside skins of the shelter. Threaded inserts shall be in accordance with MS27130 and shall be installed in the mounting members. The head of the threaded inserts shall not protrude more than 0.062 inch (1.57mm) from the shelter skin. The installed inserts for 0.3125 inch (7.94mm) bolts shall withstand an axial load of 1,000 pounts (454kg) followed by a torque load of not less than 40 lbs. in. (4.5 Nm). Unless otherwise specified (see 6.2f), inserts shall be .3125-18UNC.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

3.15 Soldering. Solder and soldering shall be in accordance with requirement 5 of MIL-STD-454.

3.16 Welding. Welding shall be in accordance with requirement 13 of MIL-STD-454.

3.17 Cleaning. After fabrication, parts, and assembled units shall be cleaned in accordance with requirement 9 of MIL-STD-454.

3.18 Light tightness. When tested as specified in 4.6.6 there shall be no evidence of difference in density of two groups of film.

3.19 Environmental. The darkroom and major components shall withstand individual environmental conditions specified herein, or combinations thereof, without damage, malfunction, or operational degradation when tested as specified in 4.6.8.

3.19.1 Temperature.

3.19.1.1 Low. During testing as specified in 4.6.8.1.1 the equipment shall meet the following requirements:

Paragraph	Measurement	Performance
3.19	Performance	Full specification requirements

After testing, the equipment shall meet full specification performance for these measurements.

3.19.1.2 High. During testing as specified in 4.6.8.1.2 the equipment shall meet the following requirements:

Paragraph	Measurement	Performance
3.19	Performance	Full specification requirements

After testing, the equipment shall meet full specification performance for these measurements.

3.19.2 Humidity. The Printer shall exhibit no physical damage, such as corrosion, rust, blistering, swelling or deterioration of parts and materials, and shall meet the requirements of 3.19 during the operating times specified in 4.6.8.2

3.19.3 Altitude (elevation). The equipment shall meet full specification performance for the following measurements both during and after testing as specified in 4.6.8.3.

Paragraph	Measurement
3.19	Performance

3.19.4 Rain. When tested as specified in 4.6.8.4 there shall be no evidence of leakage into the shelter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

3.19.5 Fording. When tested as specified in 4.6.8.5, there shall be no evidence of water penetration on the interior of the shelter.

3.19.6 Military truck transport. The darkroom, mounted on a military vehicle shall withstand the vibrations normally induced during off-road or cross-country transportation.

3.20 Stowage. All loose items shall be capable of being secured during shipment and shall remain in place during the transportability test.

3.21 Interchangeability. Like units, assemblies, subassemblies and replaceable parts shall be in accordance with requirement 7 of MIL-STD-454.

3.22 Electromagnetic interference characteristics. The equipment shall meet the emanation and susceptibility limits of MIL-STD-461 for class IIB. If the contractor considers it desirable to modify the equipment properly in order to facilitate conformance with the specified emanation and susceptibility limits, he shall obtain approval from the contracting officer before proceeding with the modification.

3.23 Finish, protective. Equipment shall be given protective finish in accordance with MIL-F-14072. This includes finish of hardware, such as handles, hinges, screws and necessary touch up after mounting. The final paint film shall be in accordance with the standard sample.

3.24 Marking. Marking shall conform to MIL-M-13231.

3.24.1 Visibility. Wherever practicable parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.24.2 Serial numbers. Each darkrooom shall be serial numbered.

3.24.3 Special marking. Labels and plates shall be provided and located in accordance with the standard sample.

3.24.4 Aircraft loading data plate. The aircraft loading data plate for the bare shelter shall be replaced with an aircraft loading plate conforming to drawing SC-D-82768.

3.25 Workmanship. The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Inspections covered by subsidiary documents (see 4.4).
c. Quality conformance inspection (see 4.5).
d. Packaging inspection (see 4.7).

4.3 First article.

4.3.1 First article units. When required, and as specified in the contract, the contractor shall furnish first article units of the complete darkroom.

4.3.2 First article inspection. The first article inspection shall consist of first article inspection specified in table I, the inspection specified in subsidiary documents covering the items listed in 4.4, and the inspections specified for group A, group B, and group C (see tables II, III, and IV respectively). The first article inspection will be performed normally as follows: (1) vehicular transport, (2) watertightness, (3) fording, and (4) electrical; other first article inspection may precede, follow, or be interspersed between the forgoing. After completion of group C inspection, conforming units shall be reinspected and shall pass all group A inspections. Failure of any inspection shall be cause for rejection of the first article.

TABLE I. First article inspection.

Inspection	Requirement paragraph	Test method paragraph
Fording	3.19.5	4.6.8.5
Transportability	3.19.6	4.6.8.6
Electromagnetic interference	3.22	4.6.9

4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirements specified in the contract.

Item	Where required
Shelter, electrical equipment S-280( )/G	3.5
Air conditioner	3.8.2
Processing machine, photographic film and paper EH-29( )	3.9.1
Printer, projection, photographic EN-36( )	3.9.2
Printer, contact, photographic EN-100( )	3.9.3
Printer, contact, photographic, continuous EN-6( )	3.9.4
Plotting table, photographic film AR-92( )	3.9.5
Film and paper rewinder	3.10
Finish	3.23
Marking	3.24

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.5 Quality conformance inspection before packaging. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract shall be inspected for conformance to the inspections specified in table II. Discrete lots shall be formed from units that pass inspection. Factors of lot composition not defined herein, or in the contract, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and acceptable quality levels (AQL) indicated in table II.

TABLE II. Group A inspection.

Inspection	Requirement paragraph	Test paragraph	AQL (percent defective) Major	Minor
Visual and mechanical	3.25	4.6.10	4.0	15.0
Breakdown voltage	3.7.4	4.6.2.1	1.5	1/
Electrical	3.7	4.6.2	1.0	1/
Light-tightness	3.13,3.18	4.6.4, 4.6.6	1.0	1/
Operational	3.8 through 3.10	4.6.3	1.0	1/

1/ All defects are considered major.

4.5.1.1 Order of inspection. Group A inspection shall be performed in any order that is satisfactory to the Government.

4.5.2 Group B inspection. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. This inspection shall conform to table III and to the special inspection levels of MIL-STD-105, table I.

4.5.2.1 Group B sampling plans. Group B AQL shall be as shown in table III and the inspection level shall be S-4 for each inspection.

4.5.2.2 Order of inspection within group B. Group B inspection shall be performed in any order which is satisfactory to the Government.

TABLE III. Group B inspection.

Inspection	Requirement paragraph	Test paragraph	AQL (Percent defective)
Interchange-ability	3.21	4.6.7	6.5
Plumbing	3.6	4.6.1	
Pressure	3.6.1	4.6.1.1	
Drainage	3.6.3	4.6.1.2	4.0 for
Pump	3.6.4	4.6.1.3	the group
Controls	3.6.6 through 3.6.9	4.6.1.4	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed group A and group B inspections. The inspection shall consist of the inspections specified in table IV.

TABLE IV. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
Elevation	3.19.3	4.6.8.3
Moisture-resistance	3.19.2	4.6.8.2
Temperature	3.19.1, 3.19.2	4.6.8.1, 4.6.8.2
Threaded inserts	3.14.2	4.6.5
Watertightness	3.19.4	4.6.8.4
Electrical resistance	3.7.2	4.6.2.2

4.5.3.1 Sampling for inspection. One sample unit shall be selected at random for all inspections in table IV from every 50 or fraction thereof produced. The first sample selected shall be at the start of the contract from the first quality conformance inspection production lot.

4.5.3.2 Order of inspection within group C. Group C inspection shall be performed in any order which is satisfactory to the Government.

4.5.3.3 Group C failures. Actions required relative to group C failures shall be as specified in the contract (see 6.2g).

4.5.4 Reinspection of conforming group B and group C sample units. Unless otherwise specified, sample units which have been subject to and passed both group B and group C inspection may be accepted on the contract provided all damage is repaired and the sample units are subject to and pass group A inspection.

4.6 Methods of inspection.

4.6.1 Plumbing.

4.6.1.1 Piping pressure. All outlets in the piping system shall be closed and the system subjected to 90 psi (621 kPa) pressure. The filter shall be removed or by-passed during this test. Nonconformance to 3.6.1 shall constitute failure of this test.

4.6.1.2 Drainage. With the shelter in a level position and filled with water or chemical solutions, attach a 50 foot (15m), 0.75 inch (19mm) inside diameter (ID) hose to the drain. Position the discharge end of the hose not more than 2 feet (0.61m) lower than the floor of the shelter and drain by gravity all water of chemical solutions. Nonconformance to 3.6.3 shall constitute failure of this test.

4.6.1.3 Pump. Using a water source located not less than 20 feet (6m) below and 100 feet (30m) from the pump through a 0.750 inch (19mm) ID heavy duty flexible hose, measure the constant discharge from the hot and cold faucets while pumping. Nonconformance to 3.6.4 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.6.1.4 Controls. The following controls shall be tested to determine compliance with the specified requirements:

Control	Requirement paragraph
Pressure regulator	3.6.6
Hot water thermostat	3.6.7
Liquid level switch	3.6.8
Pressure relief valve	3.6.9

Inability of the controls to meet the requirements as specified above shall constitute failure of this test.

4.6.2 Electrical. Electrical continuity of wiring and other connections shall be verified by a point-to-point test. Nonconformance to 3.7 shall constitute failure of this test.

4.6.2.1 Breakdown voltage. A voltage of twice the rated voltage plus 1,000 volts shall be applied for one minute between all shelter wiring and adjacent conductors not in the circuit, and between all shelter wiring and ground. The test potential may be direct current or peak alternating current of the same value. Nonconformance to 3.7.4 shall constitute failure of this test.

4.6.2.2 Electrical resistance. The power connector shall be immersed in water for 24 hours. After immersion the connector shall be tested and electrical resistance measured. Nonconformance to 3.7.2 shall constitute failure of this test.

4.6.3 Operational inspection.

4.6.3.1 Major components. Prior to installation in the shelter and at specified times during the environmental testing phase of the darkroom, the following equipment shall be subjected to the performance inspection specified in their applicable specification:

Equipment	Specification
Processing Machine, Photographic Film and Paper EH-29( )	MIL-P-55159
Printer, Projection, Photographic EN-36( )	MIL-P-55157
Printer, Contact, Photographic, EN-100( )	MIL-P-55594
Printer, Contact, Photographic, Continuous EN-6( )	MIL-P-26767

4.6.4 Light-tight container. A strip of photographic film having as ASA rating of 100 shall be placed in the light-tight container and all of the white lights in the shlter turned on for a period of 10 minutes. The film shall then be developed along with another strip of the same film which has been kept in total darkness. The two strips shall be examined for any difference in density using a standard laboratory densitometer. Nonconformance to 3.13 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.6.5 Threaded insert. Install an eyebolt in the insert to be tested. Attach a turnbuckle and a dynamometer to the eyebolt and anchor the dynamometer to a similar insert or other suitable device in the opposite panel of the shelter. Tighten the turnbuckle to obtain a 1,000 pound (454kg) tensile load on the test insert. After the load has been removed, install a bolt in the threaded insert and apply a torque wrench to the bolt to obtain a load of not less than 40 lbf. in. (4.5 Nm). Nonconformance to 3.14.2 shall constitute failure of this test.

4.6.6 Light-tightness. The darkroom shal be tested for light leaks as follows:

a. The darkroom shall be prepared for operation by uncovering all openings. The darkroom shall be subjected to an intensity of light duplicating approximately the direct sunlight at noontime.
b. Photographic film having an ASA rating of 100 shall be divided into two groups. One group shall be placed uncovered, on each major component and work area for a period of 30 minutes. The other group shall be placed in total darkness.
c. After the 30-minute period, both groups of film shall be developed simultaneously using normal processing procedures, and then examined for any difference in density, using a standard laboratory densitometer. Any fogging due to light leak shall constitute failure of this test.

4.6.7 Dimensional interchangeability. The dimensions listed below shall be measured to determine compliance with the physical interchangeability requirements of 3.21. When a dimension is not within specified limits, it shall be considered a major defect.

a. Location of threaded inserts.
b. Location of hinges.
c. Location of fasteners.

4.6.8 Environmental.

4.6.8.1 Temperature test.

4.6.8.1.1 Low temperature. The equipment shall be tested in accordance with method 502.1, procedure I, of MIL-STD-810. The storage temperature shall be -60°F (-51°C) and the operating temperature shall be 40°F (4.4°C). Prior to and after testing the equipment shall meet full specification performance for the measurements specified in 3.19.1.1.

4.6.8.1.2 High temperature. The equipment shall be tested in accordance with method 501.1, procedure II, of MIL-STD-810. The operationg temperature shall be 115°F (45°C). Prior to and after testing the equpment shall meet full specification performance for the measurements specified in 3.19.1.2.

4.6.8.2 Humidity. The printer shall be subjected to the humidity test specified in method 507.1, procedure II, of MIL-STD-810, with test measurements during Step 4 and the last 5-hour period of the fifth 48-hour cycle in Step 6. Inability of the printer to meet the requirements of 3.19.2 shall constitute failure of this test.

4.6.8.3 Altitude (elevation). The equipment shall be tested in accordance with method of 500.1, procedure I, MIL-STD-810. Prior to and after testing the equipment shall meet full specification performance for the measurements specified in 3.19.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.6.8.4 Watertightness. Darkroom without the use of any external sealing, caulking, taping, etc., shall be subjected to the following test:

Each of the five exposed surfaces of the darkroom shelter, with covers closed, shall be sprayed with water from nozzles, model G29SQ (or GG29SQ) as made by Spraying Systems Co., Bellwood, Illinois, or equal, pointing directly at the surface and arranged to provide uniform coverage of the surface. Each end panel shall be tested using 9 nozzles, each side using 15 nozzles, and the roof using 6 nozzles. Each nozzle shall operate at a pressure of 40 psig (276KPa) at the nozzle, and shall be located 19 inches (483mm) from the panel with the exception of the roof nozzles which shall be at a distance of 32 inches (813mm). Each panel shall be sprayed for a period of 40 minutes. (If the watertightness test is to be followed by the fording test, 4.6.8.5, the bottom row of nozzles can be eliminated.) See figures 5 and 6 of MIL-S-55286 for nozzle location.

4.6.8.5 Fording. The shelter, with all openings closed, shall be immersed for depth of 21 inches (533mm) above the bottom of the skids for a period of 2 hours. When the shelter is opened, after the immersion period, there shall be no evidence of water penetration inside the shelter.

4.6.8.6 Transportability. The darkroom shall be tested as specified in method 514.2, procedure XIII of MIL-STD-810.

4.6.8.7 Failure criteria. Nonconformance to 3.19 shall constitute failure of these tests.

4.6.9 Electromagnetic interference characteristics. The darkroom shall be tested in accordance with MIL-STD-462 for electromagnetic interference. The test shall be conducted with the doors closed. Nonconformance to 3.22 shall constitute failure of this test.

4.6.10 Visual and mechanical inspection. The darkroom shall be examined for the defects listed in MIL-STD-252 and table V.

TABLE V. Classification of visual and mechanical defects.

Classification	Defects
Major	1. Any item not in accordance with the standard sample, when applicable. 2. Shelter filmport, jack pads, interior painting, openins, or floor covering not as specified. 3. Sink dimensions or funnel drain not as specified. 4. Dust caps missing. 5. Tie down straps missing. 6. Sharp projections on equipment. 7. Nonconformance of labels and marking to equipment. 8. Punctures in interior skin of shelter. 9. Missing or inoperative component. 10. Foreign material which affect operation. 11. Tight wiring with possibility of breaking.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the currentversion before use.

TABLE V. Classification of visual and mechanical defects (cont'd).

Minor	1. Cracked metal not affecting use. 2. Tight wiring with no possibility of breaking. 3. Useless holes or slots. 4. Foreign material in equipment not affecting operation. 5. Scratches or abrasions on finish.

4.7 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First article inspection of packaging.
b. Quality conformance inspection of packaging.

4.7.1 First article inspection of packaging. Unless otherwise specified in the contract, first article inspection of packaging shall be in accordance with the unit pack design validation requirements of MIL-P-116.

4.7.2 Quality conformance inspection of packaging.

4.7.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.7.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B quality conformance inspection requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.7.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in table VI, "Packing inspection provisions." Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and acceptable quality level of 4.0 percent defective.

TABLE VI. Packing inspection provisions.

No.	Characteristic	Method of inspection
101	Intermediate container not as specified.	Visual
102	Improper closure of intermediate container.	Visual
103	Shipping containers not in accordance with specification.	Visual
104	Excessive cube.	Visual
105	Improper blocking and bracing.	Visual
106	Closure not in accordance with specification.	Visual
107	Weight and size exceed container limitations.	Weight & measure.
108	Strapping not in accordance with specification, incorrectly applied, or omitted.	Visual
109	Marking omitted, incorrect or illegible.	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be as specified in MIL-S-55507.

6. NOTES

6.1 Intended use. Darkroom, Photographic, Portable ES-38( ) is intended to be used as a complete self-contained mobile photographic processing laboratory capable of processing and printing 70mm, 5-inch, and 9.5 inch roll film.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When a first article is required for inspection and approval (see 3.2).
c. When a standard sample shall be lent to the contractor (see 3.3).
d. When construction of the darkroom shall not conform to the standard sample (see 3.3).
e. Time frame required for submission of statement of treatment to the contracting officer (see 3.4.2.5).
f. When threaded insert size shall be other than specified (see 3.14.2).
g. Group C failure action required (see 4.5.3.3).
h. Level A or B preservation and packing (see section 5).
i. When first article inspection rough handling tests are not required.
j. When first article packaging inspection test reports require acquisition activity approval prior to production unit packing.

6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for number of samples, examination, test, and approval of first article.

6.4 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design: for example: ES-38W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract (see 1.1).

6.5 Power supply. Electrical power to Darkroom, Photographic, Portable ES-38( ) shall be supplied by utilizing Generator Set, Trailer Mounted PU-406( )/M.

6.6 Level B preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage and handling.

6.7 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ER

Preparing activity:
Army - ER
Project 6780-A013

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

FFICIAL BUSINESS
ENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON, D.C.

POSTAGE WILL BE PAID BY DEPARTMENT OF THE ARMY

HQ. ERADCOM
Combat Surveillance & Target Acquisition
ATTN: DELCS-PE(J. Moran)
Fort Monmouth NJ 07703

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-55162B(ER)	Darkroom, Photographic, Portable ES-38( )

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426 EDITION OF 1 OCT 76 IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.